EET202

HINES & TRANSFORMERS

MODULE: 4

Percentage and Per unit Impedance, Regulation, Parallel Operation of Transformers

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Syllabus

ers –constructional details, principle of operation, EMF equation,
convention, magnetising current, transformation ratio, phasor
no load and on load, equivalent circuit, percentage and per unit
egulation. Transformer losses and efficiency, condition for
/A rating. Testing of transformers– polarity test, open circuit test,
npner's test – separation of losses, all day efficiency.Parallel
se transformers– numerical problems

tage and Per-unit values

e of any quantity is defined as the ratio of actual to the base or reference value in the same unit.

onverted into per unit quantity by dividing the the chosen base value of the same dimension.

e is dimensionless.

is expressed in percentage, it can be called .

$$Unit\ Value = \frac{Actual\ Value}{Base\ Value}$$

$$tage\ Value = Per\ Unit\ Value \times 100$$

ge and Per-unit Impedance

ted voltage

ted current

nit Impedance = $\dfrac{\text{Actual Impedance}}{\text{Base Impedance}}$

ase Impedance = $\dfrac{\text{Base Voltage}}{\text{Base Current}} = \dfrac{V_1}{I_1}$

nit Impedance = $\dfrac{Z_{01}}{V_1 / I_1} = \dfrac{I_1 Z_{01}}{V_1}$

npedance, %Z = $\dfrac{I_1 Z_{01}}{V_1} \times 100$

ge and Per-unit Impedance

nit Resistance = $\dfrac{r_{01}}{V_1 / I_1} = \dfrac{I_1 r_{01}}{V_1}$

esistance, %r= $\dfrac{I_1 r_{01}}{V_1} \times 100$

nit Reactance = $\dfrac{x_{01}}{V_1 / I_1} = \dfrac{I_1 x_{01}}{V_1}$

eactance, %x = $\dfrac{I_1 x_{01}}{V_1} \times 100$

Voltage Regulation

is loaded with a constant primary voltage, the secondary th increase in load.

resistance and leakage reactance.

f the change in secondary terminal voltage from no-load to entage of full-load secondary terminal voltage is called

tter the transformer, because a good transformer should oltage as constant as possible under all load conditions.

$$ltage\ regulation = \frac{V_{2\_NL} - V_2}{V_2}$$

$$\frac{r_{01} \cos\varphi \pm I_1 x_{01} \sin\varphi}{V_1} \times 100$$

$$\frac{r_{02} \cos\varphi \pm I_2 x_{02} \sin\varphi}{V_2} \times 100$$

+ : Lagging pf load

– : Leading pf load

Operation of Transformers

oad in excess of the rating of an existing
nd transformer may be connected in parallel.

tisfied while paralleling:

of the transformer should be suitable for the
tage and frequency.

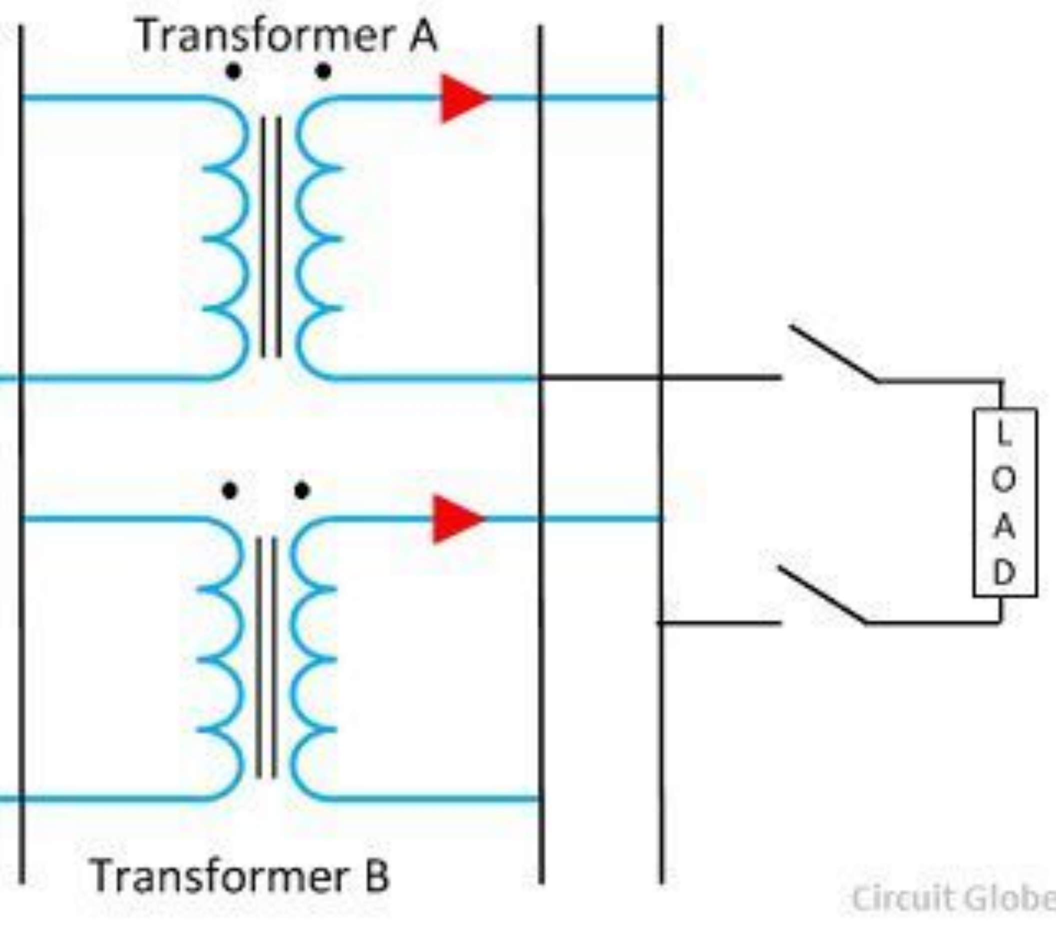
should be properly connected with regard to

g of primary and secondary should be identical. In
sformers should have same transformation ratio.

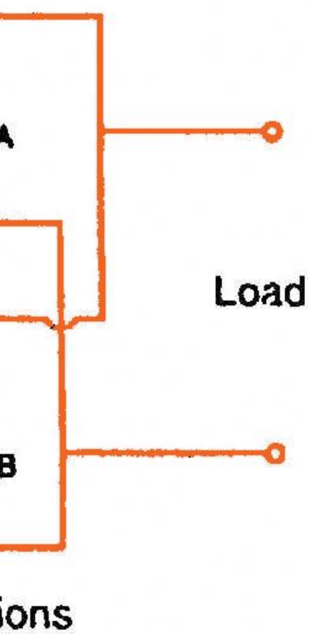
dances should be equal

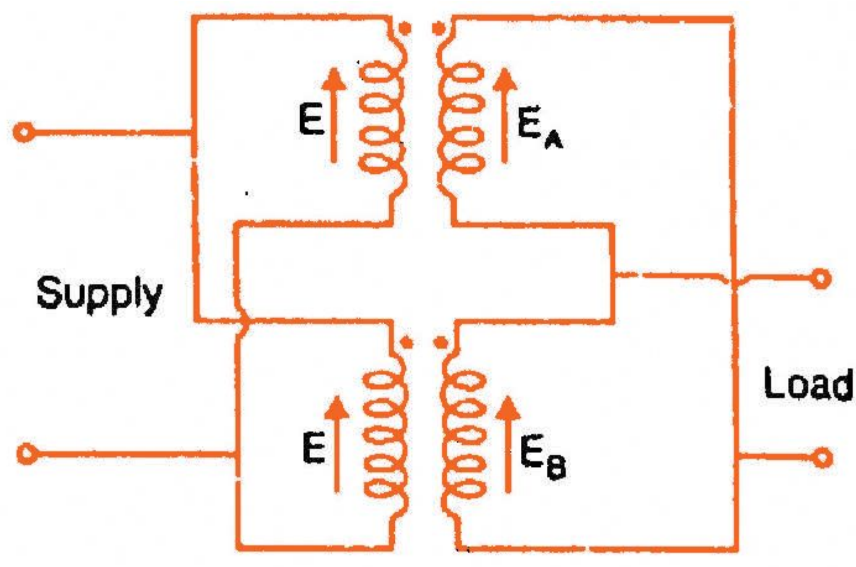
valent leakage reactance to equivalent resistance
or all the transformers.

Operation of Transformers



Operation of Transformers





Wrong connections
(ii)